THE UNIVERSITY OF ARIZONA
College of Information Science

INFO – 698: CAPSTONE

Final Report
on
Independent Research Study for Domain Adaptation

FINE TUNING FOUNDATIONAL MUSIC MODEL FOR CARNATIC AUDIO CONTINUATION

Submitted by

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1. Executive Summary

This capstone research study develops, evaluates and documents a culturally adapted variant of Meta AI's open-source MusicGen [1] foundation music model that operates on EnCodec residual vector-quantized (RVQ) audio tokens specialized for Carnatic music continuation. Carnatic music, one of South India's two principal classical traditions, is characterized by raga-grammar melody, microtonal ornamentation (gamakas) and cyclic tala structures, and is systematically underrepresented in the Western-dominated corpora that pretrain modern foundation music models. Working entirely within an audio-to-audio continuation framework, this study uses Low-Rank Adaptation (LoRA) to fine-tune MusicGen-small on approximately ninety-six hours of cleaned audio derived from the licensed dataset created by CompMusic, UPF named Indian Art Music Raga Recognition Dataset (audio) [2].

Domain adaptation is evaluated using a hybrid framework combining cross-entropy/ perplexity training metrics, four boundary-continuation cosine distances on Mel, MFCC, Chroma and Onset features, and a user perceptual listening study with both 5-point Likert and choice pairwise tasks. Pairwise outcomes are tested with one-sided Holm-corrected binomial tests at $\alpha = 0.05$ and the fine-tuned model is preferred at statistically significant levels on every perceptual question. The full pipeline, evaluation artefacts and the listening study are released as a reproducible GitHub repository together with a hosted Streamlit web survey.

2. Literature Review

2.1 About Carnatic Music

Carnatic music is one of the world's oldest and most deeply rooted musical traditions, predominantly cultivated in South India. [3]. Unlike Western tonal music, which centers on harmony and equally tempered scales, Carnatic music is fundamentally melodic and microtonal.[4]. Its expressive vocabulary is built around the raga, a melodic framework that prescribes a set of swaras (notes), characteristic phrases (sancharas) and ornamental movements (gamakas) that govern how those swaras must be approached and embellished. [5]. The 72-melakarta system organizes parent ragas into a structured grid, and thousands of derivative janya ragas inherit their grammar. [6].

2.2 Evolution of Music Generation Models

Music generation has evolved significantly over the last decade. Early neural generative approaches proposed by (Oord, AVD. et al., 2016) [8], introduced autoregressive waveform modelling. However, raw audio modelling at scale was computationally expensive, as further discussed by (Dieleman, S. et al., 2018) [9].

Subsequent systems transitioned toward symbolic modelling using RNNs and Transformers. A Survey on Deep Learning for Symbolic Music Generation by (Ji, S. et al., 2023) [10] documents progression from LSTM-based melody generation to attention-based architectures like Museformer [11]. More recently, discrete token-based modeling enabled language model style approaches to music generation. AudioLM introduced by scientists in Google DeepMind through (Borsos, Z. et al., 2023)[12] demonstrated hierarchical token modeling for long-form audio continuation through semantic and acoustic tokens. Further, MusicLM by (

Agostinelli, A. et al., 2024) [13] introduced text-conditioned music generation. Diffusion-based approaches such as AudioLDM by (Liu, H. et al., 2023) [14] and AudioLDM 2 by (Liu, H. et al., 2024) [15] further expanded text-to-audio capabilities. Subsequently, the emergence of foundation music models, challenges of manually annotated dataset availability and lack of specialized music expertise for music comprehension were surveyed by (Li, W. et al., 2024) [16].

Recent survey by (Mitra, R. et al., 2025) [17] presents a systematic review of recent advances in music generation. While the paper underscores the emergence of diffusion models and its training methods, it also evaluates and identifies the best performing models to be GANs and Language Models.

2.3 From Text-to-Audio to Audio-to-Audio

Most recent research including Mustango, Instruct-MusicGen, ChatMusician and Video-Guided Text-to-Music Generation [18-21] focuses on text-to-music systems. However, these paradigms require large, paired prompt-audio datasets. These types of datasets are available for western music, while for culturally specific traditions such as Carnatic music, prompt-labelled corpora are scarce. This is identified in the recent work of (Ravikumar, PT., 2024) [22] called

DAIRHuM that highlights challenges in aligning AI models with human judgments in Carnatic music due to scarcity of annotated ground truth and culturally contextual labels, contrasting with better-supported Western datasets. Also, the survey by (Zhao, Y. et al., 2025) [23] identifies the core challenges of data scarcity, representational biases, and long-term coherence involved in text to audio generation.

A recent empirical demonstration of this scarcity problem is (Mehta, A. et al., 2025) [24], which fine-tuned MusicGen and Mustango on Hindustani Classical and Turkish Makam by manufacturing prompt templates from Dunya and Saraga metadata. The authors report only moderate improvements and observe that their adapted models lose performance on Western genres, evidencing the non-triviality of cross-cultural text-to-music adaptation when authentic prompt-labelled corpora are unavailable.

Audio-to-audio continuation and hierarchical token-based generative models including AudioLM [12] and MusicGen [1], allow domain adaptation using pure audio without requiring text annotations. This makes audio continuation particularly suitable for culturally rich audio datasets rather than prompt-labelled datasets.

2.4 Cultural Bias in Music Generation

While survey by (Li, W. et al., 2024) [16] spoke about the challenges of manually annotated dataset availability and lack of specialized music expertise for music comprehension, the issue of cultural imbalance is explicitly expressed in (Mehta, A. et al., 2025) [24] and (Roca, RB. et al., 2025) [25]. These works highlight that music generation systems trained predominantly on Western datasets underperform on non-Western traditions. Commentary on Saraga Open Dataset [7] by (Pearson, L., 2021) [26] presents additional modelling challenges in Carnatic music like Microtonal ornamentations (gamakas), Long improvisational alapana and Cyclic tala structures. Though existing phrase-based raga recognition work using vector space modeling, introduced in (Gulati, S. et al., 2016) [27] leverages the audio dataset [2] that provide structured and authentic Carnatic recordings, this dataset has not been used for foundation model domain adaptation.

2.5 Fine-Tuning and Transfer Learning

Fine-tuning strategies for large language models have evolved significantly alongside the development of foundation architectures. A review work published by (Huang, Z. et al., 2025) [28], establishes the broader methodological shift toward more efficient adaptation strategies, which outlines the progression from full-parameter updating toward structured and parameter-efficient adaptation techniques designed to preserve pretrained knowledge while specializing for domain-specific tasks. This is also evident from the papers that introduced AudioLM [12] and MusicLM [13].

In the context of music foundation models like MusicLM and MERT, this evolution is explicitly examined in (Ding, Y. et al., 2024) [29]. Their work categorizes modern fine-tuning strategies into three principal classes: (1) full fine-tuning, (2) adapter-based methods, and (3) prompt-based or reparameterization-based methods. Full fine-tuning modifies all weights and provides maximum flexibility but incurs high computational cost and risks catastrophic forgetting. In contrast, parameter-efficient transfer learning (PETL) techniques update only a small subset of parameters through lightweight modules inserted into the pretrained backbone. These methods preserve general musical knowledge learned during pretraining while adapting to specific musical domains with significantly reduced computational overhead. The study emphasizes that adapter-based approaches are particularly effective for domain-specific music tasks where training data is limited.

Lightweight adaptation strategies are further explored in (Tsai, FD. et al., 2024) [30] for music editing abilities of Text-to-Music generative methods. Rather than modifying the full backbone model, this work introduces attention-based adapter modules that integrate audio-derived conditioning features into a pretrained generator. Only the adapter components are trained, while the core model remains frozen.

A recent application of PEFT to cultural-domain adaptation is (Mehta, A. et al., 2025) [24], which inserts Pfeiffer-style adapter modules into MusicGen and Mustango and fine-tunes them on Hindustani Classical and Turkish Makam. The work reports moderate gains on the target traditions but also observes catastrophic forgetting of Western genres, illustrating the trade-off between adapter capacity and pretrained knowledge preservation that any cultural fine-tuning strategy must navigate.

The design considerations of such adapter mechanisms are examined in (Mehta, A. et al., 2025) [31]. This study evaluates architectural choices in adapter design and highlights trade-offs between convolution-based adapters, which capture local musical features, and transformer-based adapters, which better preserve long-range musical dependencies. The study also shows that adapter size and placement significantly influence adaptation performance in low-resource music settings.

Beyond supervised fine-tuning, reinforcement learning has also been proposed as an adaptation mechanism in (Guo, X. et al., 2022) [32]. The authors introduce a reinforcement learning framework where a pretrained transformer model is further optimized using reward signals reflecting musical quality and structural constraints. This approach allows optimization towards perceptual or rule-based objectives that may not be directly captured by likelihood-based training. Reinforcement learning thus represents a complementary fine-tuning strategy aimed at aligning generative outputs with musical preferences or domain constraints.

Ultimately, the work by (Lu, SC. et al., 2026) [33] introduces a structured symbolic representation designed to enhance the fine-tuning of large language models for music generation by transforming raw music data into a notation format that captures hierarchical musical structure. Instead of fine-tuning directly on raw waveform or token streams, YNote converts music into a high-level representation that preserves rhythmic, harmonic, and structural features, allowing pretrained language models to better learn musical dependencies during adaptation. This intermediate representation serves as an efficient mediation layer, reducing the complexity of the fine-tuning task and enabling more effective transfer from general pretrained models to music generation domains with structured compositional requirements.

While the literature [28-33] demonstrates that large pretrained models can be effectively adapted to different tasks and domains through fine-tuning using advance parameter efficient methodologies, adapter-based techniques or a combination of both, and YNote, there is no study that has systematically fine-tuned a foundation music model specifically for Carnatic audio continuation using authentic datasets.

2.6 Evaluation in Music Generation

A critical dimension in music generative research is evaluation. Unlike tasks with discrete correctness (e.g., classification), music generation quality is inherently subjective and multi-faceted. Recent surveys on this subject highlight the need for structured evaluation frameworks in generative music research.

The survey by (Lerch, A. et al., 2025) [34] provides a structured taxonomy of evaluation methods for music generation systems and categorizes them into quantitative (objective) and qualitative (subjective) approaches. On the quantitative side, the survey discusses signal-level similarity metrics such as spectral convergence, Mel-spectrogram distance, and reconstruction error, as well as distributional metrics like Fréchet Audio Distance (FAD) and embedding-based similarity measures. It also highlights feature-based evaluation methods where pitch statistics, rhythmic descriptors, tonal stability, and structural repetition are analyzed numerically. On the qualitative side, the survey outlines perceptual listening studies including Likert-scale ratings (e.g., realism, coherence, creativity), Mean Opinion Score (MOS), pairwise preference testing, and MUSHRA-style evaluations. It emphasizes that music evaluation cannot rely solely on objective metrics because musicality and stylistic authenticity require human perceptual validation. Importantly, the survey also advocates for hybrid evaluation frameworks combining objective feature metrics with structured human listening experiments.

The work by (Li, C. et al., 2025) [35] proposes a benchmark dataset explicitly designed to standardize evaluation of text-to-music systems. The authors combine quantitative automatic metrics with qualitative expert human ratings. On the qualitative side, expert musicians provide structured evaluations using Likert-scale scoring for attributes such as musicality, coherence, adherence to prompt, and aesthetic quality. These ratings serve as ground truth for perceptual realism. On the quantitative side, the study explores automatic metrics including embedding similarity scores, text-audio alignment measures, and acoustic feature-based comparisons. The key contribution is correlational analysis between automatic metrics and expert human ratings, identifying which quantitative metrics best approximate perceptual judgments.

In the paper published by (Li, J. et al., 2024) [36], the authors introduce a large-scale benchmark designed to evaluate music-related reasoning and generation capabilities in language models. The evaluation framework combines structured quantitative testing with qualitative perceptual analysis. Quantitatively, the benchmark includes accuracy-based tasks (e.g., symbolic music reasoning, genre classification, theoretical knowledge tests) and alignment scoring metrics to measure how well generated music adheres to prompts or theoretical constraints. For generative outputs, statistical measures and embedding-based comparisons are used to assess structural coherence. Qualitatively, human evaluators assess musical plausibility, structural correctness, and stylistic alignment using rating scales. The benchmark also incorporates preference-based evaluation to compare outputs across models.

(Guo, Y. et al., 2023) [37] represents a framework that integrates generative modeling with systematic evaluation. On the quantitative side, the authors analyze pitch distribution similarity, rhythmic consistency metrics, tonal interval distributions, and structural repetition patterns to compare generated outputs against real compositions. These feature-based metrics assess statistical alignment with training data. Additionally, classification accuracy is used as an indirect measure of stylistic fidelity. On the qualitative side, human listeners evaluate musicality, smoothness, and overall listening experience through structured surveys. The study demonstrates how transfer learning can preserve musical structure while improving stylistic adaptation, and it emphasizes that statistical similarity alone does not guarantee perceptual quality, reinforcing the need for listener-based evaluation.

Additionally, evaluation methods in MusicGen [1] and AudioLM [12] combine both quantitative and qualitative strategies to validate musical fidelity and perceptual quality. In MusicGen, quantitative metrics such as token-level accuracy, chord and rhythmic alignment accuracy, and statistical comparisons of generated musical features (e.g., pitch and temporal distributions) against conditioning inputs assess controllability and structural correctness.

Complementing this, the qualitative human listening tests using Likert-scale ratings to evaluate musical coherence, stylistic plausibility, and perceived adherence to controls. Similarly, AudioLM employs quantitative evaluation through language-model likelihood measures, token prediction accuracy, and acoustic similarity metrics such as Mel-spectrogram comparison and Mel-cepstral distortion to assess signal-level fidelity and continuation consistency. AudioLM also conducts qualitative human listening studies, including pairwise preference tests and Mean Opinion Score (MOS)-style evaluations, where listeners judge naturalness, smoothness of continuation, and overall audio realism.

From the references [1, 12, 34-38], it can be inferred that evaluation in music generation differs depending on the generation paradigm. In text-to-audio models, evaluation often emphasizes prompt alignment and semantic consistency, using metrics such as text–audio embedding similarity, alignment scores, and human ratings for adherence to the prompt, musicality, and aesthetic quality. In contrast, audio-to-audio continuation models prioritize signal fidelity and temporal coherence, relying more heavily on acoustic similarity metrics (e.g., Mel-spectrogram distance, Mel-cepstral distortion), likelihood-based measures, and distributional comparisons such as Fréchet Audio Distance. Across both paradigms, however, literature consistently emphasizes the necessity of hybrid evaluation frameworks that combine objective quantitative metrics with qualitative human listening studies to assess realism, coherence, and stylistic authenticity.

2.7 Identified Research Gap

From the afore indicated sections 2.2 to 2.6, three observations surface. (i) Text-to-audio systems dominate current research, but prompt-labelled dataset for culturally oriented music is unavailable and is considered a daunting and subjective task [22]. (ii) Cultural bias in foundation music models is empirically documented and only partially mitigated and bottleneck adapter PEFT on MusicGen and Mustango yields moderate gains for Hindustani Classical and Turkish Makam while inducing catastrophic forgetting of Western genres [24, 31]. (iii) Authentic licensed Carnatic audio corpora exist but are underutilized in generative foundation model adaptation.

While (Mehta, A. et al., 2025) [24] established that Pfeiffer-style adapter modules yield moderate gains for Hindustani Classical and Turkish Makam under text-conditioning, the LoRA family of PEFT methods which adapts via inserting low-rank decomposition matrices has not been evaluated for cultural domain adaptation of MusicGen.

This study addresses these gaps in the Carnatic setting. First, it adapts an unconditional audio-to-audio continuation protocol that obviates the textual prompt template manufacturing. Second, it applies LoRA to the linear projections within the autoregressive language model of MusicGen-small and fine-tunes on approximately 96 hours of cleaned Carnatic audio drawn from a licensed corpus [2], and third, evaluates the result under an unconditional continuation protocol using both boundary continuation distance metrics over Mel, MFCC, Chroma, and Onset features, and a Holm-corrected pairwise Perceptual listening study stratified by Carnatic familiarity and by instrument family.

3. Goal

The goal of this study is to design, train, and evaluate a LoRA adapted variant of MusicGen-small that, given a 10-second Carnatic audio prompt and operating under an unconditional audio-to-audio continuation protocol, produces continuations judged more Carnatic authentic than those of the pretrained baseline by both objective boundary continuation metrics and a Holm-corrected perceptual listening study, while exhibiting stable, non-overfitting training dynamics on the held out validation split.

4. Research Questions

The study is organized around three concrete research questions:

1. *RQ1 (Trainability):* Can the MusicGen-small autoregressive language model be adapted to the Carnatic domain using a small fraction of trainable parameters through LoRA, while still producing measurable improvements in token-level cross-entropy loss?
2. *RQ2 (Quantitative continuity):* Does LoRA fine-tuning improve the acoustic continuity of generated continuations across the prompt to continuation boundary, when measured using mean cosine distances over Mel, MFCC, Chroma and Onset feature vectors?

3. *<u>RQ3 (Perceived authenticity):</u>* Do human listeners, both Carnatic-familiar and Carnatic-unfamiliar prefer the fine-tuned continuations over the baseline continuations on perceptual axes of musicality, Carnatic authenticity and continuity, and is this preference statistically significant after multiple-comparison correction?

5. Benefits

This work contributes to the broader ambition of more inclusive AI systems that understand and respect diverse musical traditions. A culturally adapted foundation music model for Carnatic continuation has direct implications for: (i) cultural preservation, by establishing a reproducible audio-to-audio adaptation recipe that can be replayed for other underrepresented traditions; (ii) educational and creative tooling, where a stylistically informed continuation engine can support learners, instructors and composers exploring raga-conditioned ideation; (iii) AI research practice, by demonstrating that parameter-efficient adaptation can succeed on a small commodity-class device, thereby lowering the resource floor for cultural domain-adaptation research; and (iv) archival restoration, where significant Carnatic recordings still survive only on legacy media such as cassettes and CDs that degrade over time. A stylistically informed continuation engine offers a principled way to reconstruct missing or damaged passages while preserving the artistic register of the original recording.

6. Core function and technical approach

This section describes the end-to-end pipeline that takes raw, full-length Carnatic recordings and produces a LoRA-adapted MusicGen-small checkpoint together with a complete evaluation bundle. The pipeline is implemented in Python 3.9 against a patched Apple-Silicon-compatible build of Meta's AudioCraft library, executed inside a cursor / VS Code Jupyter kernel and accelerated through Metal Performance Shaders (MPS) on the Apple M-series GPU. The orchestration layer (pipelines.py) chains modular helper functions (helpers.py) and dataset / LoRA classes (classes.py) into ten sequential stages.

6.1 Methodology

Figure 1 summarizes the methodology adopted for Data Curation, Fine Tuning and Evaluation. The pipeline runs from data access through evaluation, with a clearly demarcated fine-tuning loop and a hybrid evaluation block at the end.

Each stage emits an artifact (CSV metadata, .wav segment, .pt token tensor, training-history CSV, evaluation .wav, plot) that the next stage consumes, allowing individual stages to be re-run independently. Reproducibility is supported by fixed random seeds (RANDOM_SEED = 42 for splits), explicit hyperparameter logging in helpers.py and complete metadata CSVs at every stage.

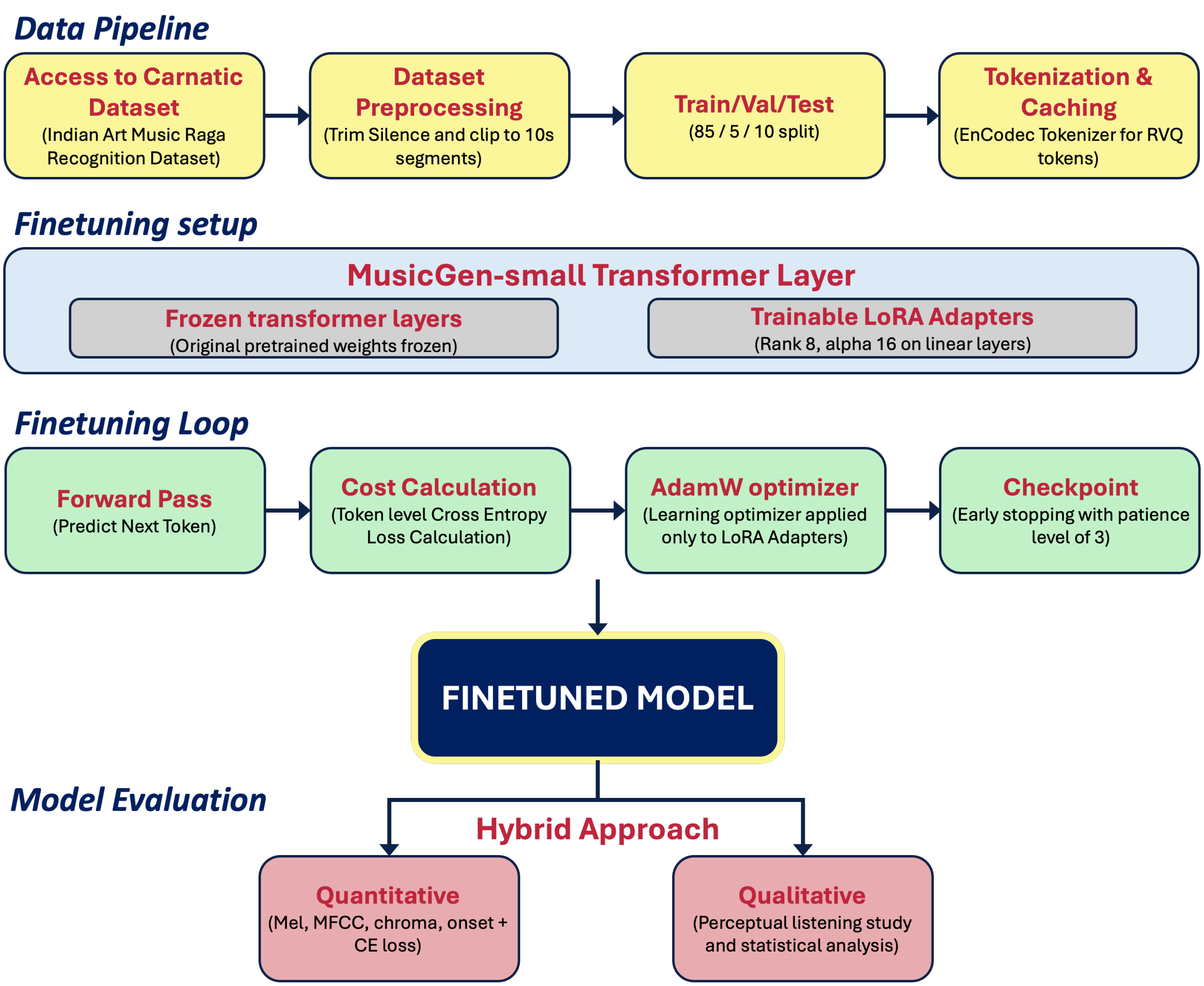

Figure 1 *Overview of Methodology*

6.2 Dataset Collection and Preparation

Several Carnatic music datasets were considered before settling on the source corpus, including community-uploaded YouTube collections, Saraga (Srinivasamurthy et al., 2021) [7] and informal Kaggle dumps, but only the Indian Art Music Raga Recognition Dataset released by Serrà et al. (2016) [2] through CompMusic, UPF provided licensed audio for research-only access through Zenodo, a substantial corpus of authentic Carnatic recordings and a structured artist/ raga directory layout. The corpus ships as MP3 files, which were converted to 32-kHz mono .wav files (the native input rate of MusicGen and EnCodec).

Each .wav file was passed through a cleaning and segmentation pipeline. Audio was loaded as a single channel, resampled to 32 kHz and trimmed of leading and trailing silence below an amplitude threshold of 0.01. Trimmed audio was split into non-overlapping 10-second clips (320,000 samples) and each clip was scored against three quality heuristics: a minimum RMS amplitude of 0.005 (rejecting near-silent clips), a maximum near-zero ratio of 0.30 (rejecting silence-heavy clips) and a minimum non-silent ratio of 0.60 (requiring at least six seconds of active sound). Clips violating any criterion were rejected. Table 1 summarizes the dataset preprocessing outcome.

Stage	Count / Value
Source recordings	426
Total 10-second segments	39,162
Segments accepted	34,400
Segments rejected (too much silence)	4,718
Segments rejected (low RMS)	44
Effective training corpus	≈ 95.6 hours of clean Carnatic audio

Table 1 *Summary of dataset preprocessing outcomes.*

6.3 Code Architecture

The codebase is organized into four orchestration layers. *classes.py* defines the custom CachedRVQDataset (a torch.utils.data.Dataset that loads pre-cached RVQ token tensors from a CSV manifest) and LoRALinear (a low-rank adapter that wraps a frozen base nn.Linear layer and adds two trainable matrices). *helpers.py* contains all reusable primitives like device/ model loading, audio cleaning, segmentation, train/ validation/ test splitting, EnCodec token caching, LoRA injection/ training/ checkpointing, baseline-vs-fine-tuned generation and boundary-continuation evaluation. *pipelines.py* composes these helpers into runnable end-to-end stages (reproducibility check, EnCodec confirmation, dataset preparation, tokenization, baseline evaluation, LoRA fine-tuning, post-fine-tuning evaluation, quantitative evaluation). *fine-tuning.ipynb* is the executable notebook that invokes these stages sequentially under the custom Python kernel.

Library	Role in the pipeline
torch / torchaudio	Model definition, training loop, audio loading and resampling.
audiocraft (Apple-Silicon fork)	MusicGen / EnCodec backbone, generate_continuation API.
transformers	Hugging Face MusicgenForConditionalGeneration sanity checks.
librosa	Boundary-continuation feature extraction (Mel, MFCC, Chroma, Onset).
scikit-learn	Recording-level train / val / test splitting (fixed seed).
pandas, numpy, scipy	Metadata storage, statistical computation, audio I/O helpers.
matplotlib	Training history and evaluation plots.
Streamlit + gspread / Google Drive API	Listening-study front-end and Google-Sheet response logging.

Table 2 *Key Python libraries used in the pipeline.*

6.4 Train/ Validation/ Test Split

To prevent leakage between splits, partitioning was performed at the recording level rather than the segment level, using scikit-learn's train_test_split with a fixed RANDOM_SEED of 42. Recordings were split 85 % train / 5 % validation / 10 % test (test taken first, then validation taken as 5/90 of the remaining 90 %). Every segment inherited the split label of its source recording. Table 3 reports the resulting counts.

Split	Recordings	Segments	No. of Hours
Train	361	28,485	≈ 79.1
Validation	22	2,168	≈ 6
Test	43	3,747	≈ 10.4

Table 3 *Recording-level train/ validation/ test splits (fixed seed = 42).*

6.5 EnCodec RVQ Tokenization

MusicGen's autoregressive language model operates on Residual Vector Quantized (RVQ) discrete codes produced by Meta's EnCodec compression model at 32 kHz. To eliminate redundant tokenization cost during training, every accepted 10-second segment was encoded once and cached to disk as a Torch payload containing the codebook tensor (shape [K codebooks × T frames]).

At training time, the custom CachedRVQDataset class loads each cached payload directly into memory, returning the long-typed code tensor without re-running the encoder. This caching design allowed the entire fine-tuning pipeline to run locally without saturating GPU memory and made each training epoch dependent only on the autoregressive language-model forward pass.

6.6 LoRA Fine-Tuning of MusicGen-Small

The pretrained baseline is facebook/musicgen-small loaded through AudioCraft's MusicGen.get_pretrained API. All base parameters of the language model are frozen at the start of fine-tuning. Adaptation is performed via Low-Rank Adaptation (LoRA). Each eligible nn.Linear layer of the LM transformer is wrapped by a LoRALinear module that retains the frozen base weight W_0 and adds two trainable low-rank matrices $A \in \mathbb{R}^{\{r\times k\}}$ and $B \in \mathbb{R}^{\{d\times r\}}$; the layer's output becomes

$$y = W_0 X + (\alpha/r) \cdot B \cdot A \cdot X.$$

Equation 1 *LoRA equation*

Following the original LoRA formulation, A is initialized with Kaiming uniform sampling and B is initialized to zero, so the adapter contributes nothing at the start of training and only develops a non-zero residual as it learns.

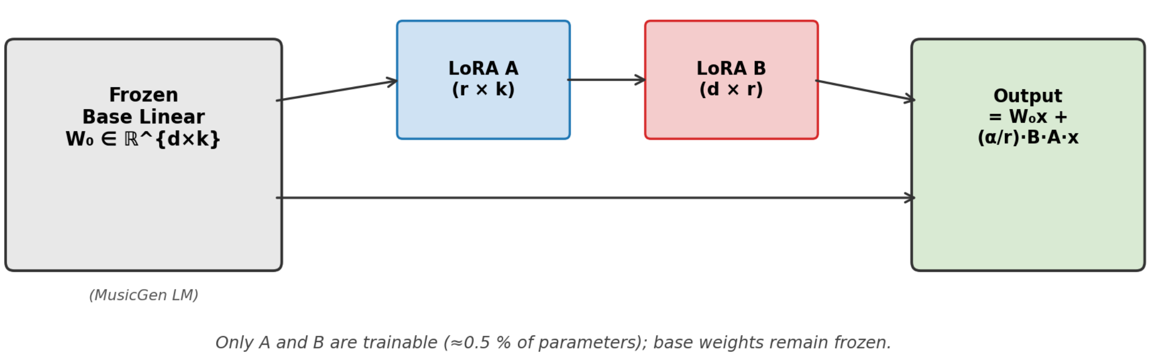

Figure 2 *LoRA adapter injection on each linear layer.*

Wrapping is applied recursively and LoRA injection is restricted to nn.Linear modules as these linear layers capture almost all the trainable parameters in a transformer. A MusicGen LM block is built from: multi-head self-attention (the Q, K, V, and output-projection that are all nn.Linear) + a feed-forward MLP (two more nn.Linear layers) + LayerNorms + residual connections + activations. The Linear layer carries most of the weights, and they're the surfaces on which adaptation happens. Therefore, adapting the linear layer is high leverage. The condition_provider modules are deliberately excluded from the LoRA wrapping, because this study performs unconditional audio continuation and does not require text-conditioning adaptation. The hyperparameters used for fine tuning are summarized in Table 4.

Hyperparameter	Value	Rationale
Baseline	facebook/ musicgen-small	Starter variant of MusicGen.
LoRA rank (r)	8	Standard low-rank choice; ≈ 0.5 % trainable parameters.
LoRA alpha (α)	16	Sets effective scaling factor α/r = 2.
LoRA dropout	0.05	Light regularization on the LoRA path.
Optimizer	AdamW	Standard for transformer LM fine-tuning
Learning rate	1×10^{-4}	Conservative LR to preserve pretrained knowledge.
Weight decay	1×10^{-4}	Regularization against large updates.
Batch size (per step)	1	MPS / RAM compatible
Gradient accumulation	8 steps	Simulates effective batch size of 8.
Max gradient norm	1.0	Gradient clipping for training stability.
Epochs	10 (early-stop patience = 3)	Best validation epoch selected.

Table 4 *LoRA fine-tuning hyperparameters for MusicGen-Small adaptation.*

The post training objective is the standard token-level cross-entropy loss on the masked output of MusicGen's compute_predictions API, computed against the cached EnCodec codes. A single dummy Conditioning Attributes (empty text description) is built on CPU and moved to the LM device per batch, which lets training in unconditional mode without invoking the live T5 text encoder on every step. Further, the post training loop performs gradient accumulation, gradient clipping, per-epoch validation, last/ best LoRA checkpoint extraction and early stopping. Per-epoch train and validation cross-entropy are also recorded. Only the LoRA matrices are saved in the checkpoint payload, keeping each on-disk checkpoint tiny.

6.7 Generation Protocol for Evaluation

Post-training evaluation generates audio Audio continuations. 50 test segments are sampled with random_state = 42 from the held-out test split; each segment is loaded as a 10-second prompt, and both the baseline pretrained MusicGen-Small and the LoRA-adapted model generate a 20-second waveform conditioned on the prompt (10 s prompt + 10 s continuation horizon) using AudioCraft's generate_continuation API on CPU for determinism. These 50 continuation pairs feed both the boundary-continuation quantitative metrics and the human listening study.

6.8 Quantitative Evaluation

Quantitative evaluation operates at two granularities. At the post training granularity, the LoRA fine-tuning loop logs token-level cross-entropy loss on both the train and validation splits at every epoch, from which we derive validation perplexity (exp of CE loss) and select the best epoch by lowest validation CE. Perplexity, defined as the exponential of the cross-entropy loss, expresses the model's predictive uncertainty as the effective number of equally likely next tokens it is choosing between at each step. We report validation perplexity alongside validation CE for three reasons: it makes the absolute scale of progress interpretable (the run drops from 66.18 to 62.84 across ten epochs), it diagnoses overfitting or saturation through divergence or flattening across epochs, and it is the standard yardstick in the audio-language-model literature, ensuring our results remain comparable to MusicGen, AudioLM and MusicLM benchmarks as well as to future scaled-up variants of this same study. At the inference granularity, we adopt boundary-continuation distance metrics that measure how acoustically continuous the generated continuation is across the prompt to continuation seam. For every continuation clip we extract a one-second boundary window (the last 1 s of the 10-s prompt and the first 1 s of the 10-s continuation), and compute four time-averaged acoustic feature vectors per window: a 128-bin Mel-spectrogram (log-power, time-averaged), a 20-coefficient MFCC (time-averaged), a 12-bin chroma-STFT vector (time-averaged) and a frame-level onset-strength envelope (truncated to common length). For each feature, we compute the cosine distance between the prompt-boundary and the continuation-boundary vector. Lower distance indicates a smoother prompt to continuation transition.

6.8.1 Training Dynamics: Cross-Entropy Loss and Perplexity

Figure 3 plots training and validation cross-entropy loss for the reported 10-epoch LoRA fine-tuning run. Both losses decrease monotonically across all ten epochs without divergence; validation loss falls from 4.1920 at epoch 1 to 4.1402 at epoch 10, with the lowest validation loss attained at epoch 10. The training loss decreases in lockstep, and the small consistent gap between train and validation suggests light, well-controlled regularization rather than overfitting on this corpus size.

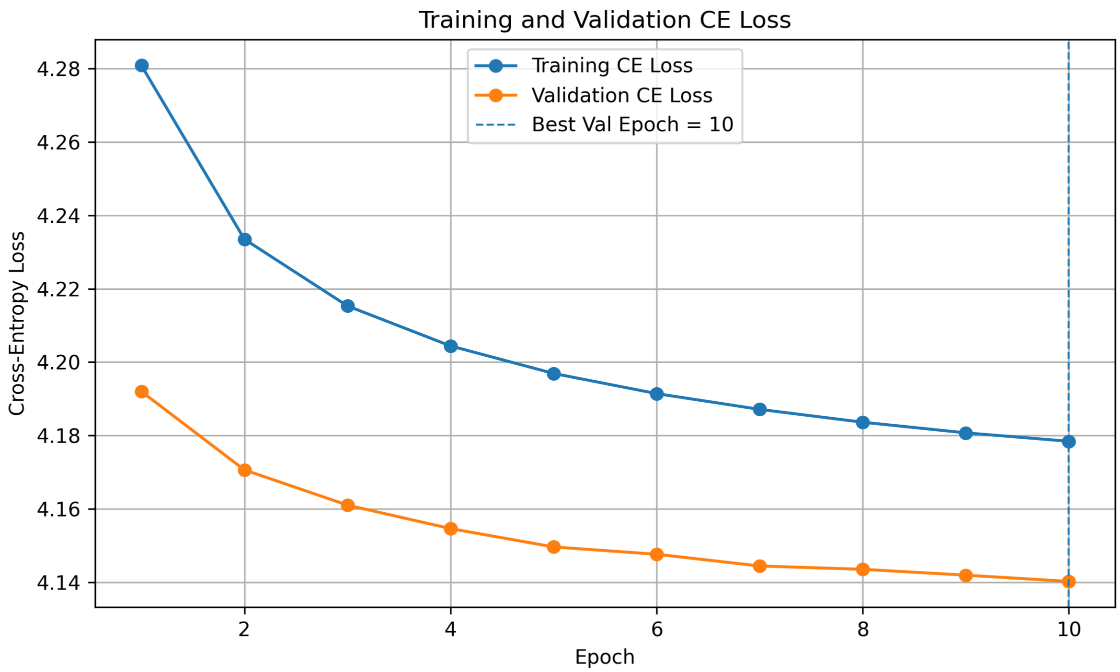

Figure 3 *Training and validation cross-entropy loss across LoRA fine-tuning epochs*

Epoch	Train CE Loss	Val CE Loss	Val Perplexity
1	4.2809	4.1920	66.18
2	4.2335	4.1706	64.78
3	4.2153	4.1610	64.16
4	4.2044	4.1546	63.75
5	4.1969	4.1496	63.43
6	4.1914	4.1476	63.30
7	4.1871	4.1444	63.10
8	4.1836	4.1435	63.04
9	4.1807	4.1419	62.94
10 (best)	4.1784	4.1402	62.84

Table 5 *Per-epoch training and validation cross-entropy loss and validation perplexity.*

Two observations follow. First, the absolute drop in validation CE (≈ 0.052 nats) is small, but is achieved by a model with only ≈ 0.5 % of its parameters trainable; this indicates that LoRA can shift MusicGen-small in the Carnatic direction without disturbing it's pretrained foundation. Second, the curve is still trending downward at epoch 10, suggesting that longer training (or a larger LoRA rank) would likely yield further improvement.

6.8.2 Boundary-Continuation Distance Metrics

The Carnatic relevance of each of the boundary metric is as in the table 6 below.

Metric	Acoustic property	Carnatic relevance
Mel-spectrogram distance	Spectral energy distribution (128 bands)	Vocal / instrumental tone continuity.
MFCC distance	Spectral envelope (20 cepstral coefficients)	Vocal-tract shaping, bowing technique, attack quality. Most sensitive to *how* a note is produced.
Chroma distance	pitch-class energies of 12 notes	12-bin grid does not natively represent Carnatic microtones or gamakas.
Onset-envelope distance	Frame-level attack strength	Closest proxy for tala-style consistency but does not judge tala correctness.

Table 6 *Carnatic relevance of boundary metrics*

Boundary-continuation cosine distances were computed for the full 3,700 valid prompt windows derived from the test split (each prompt becoming a 1-second boundary anchor against the corresponding 1-second continuation boundary). Table 6 reports baseline and fine-tuned mean distances, the absolute and percent improvement of fine-tuned over baseline, and the fraction of clips on which the fine-tuned model strictly beats the baseline. Lower distance = smoother continuation.

Boundary metric	Baseline mean	Fine-tuned mean	Δ (Baseline − FT)	% improvement
Mel-spectrogram distance	0.0062	0.0061	0.0001	1.3 %
MFCC distance	0.0252	0.0241	0.0011	4.2 %
Chroma distance	0.1358	0.1363	- 0.0006	- 0.4 %
Onset-envelope distance	0.3792	0.3746	0.0045	1.2 %

Table 7 *Mean boundary-continuation cosine distances.*

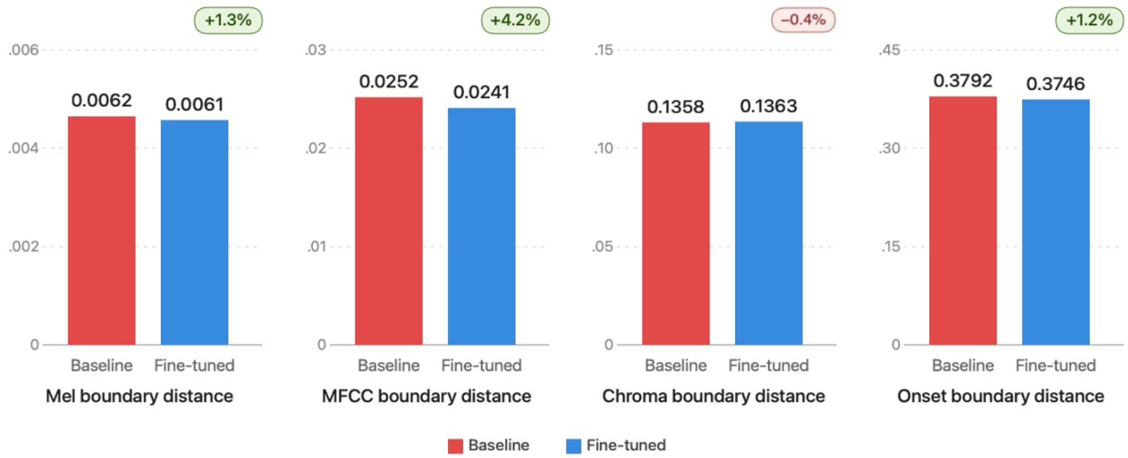

***Figure** 4 Baseline vs fine-tuned boundary distances.*

We observed three results. First, the fine-tuned model produces measurable, consistent improvements on Mel, MFCC and Onset distances (1.19–4.25 % relative reductions in mean distance), with the largest gain on MFCC (the timbral / vocal-tract feature), most sensitive to articulation. Second, Chroma distance is essentially flat (a marginal regression of 0.43 %), which is consistent with the dataset retaining many baseline harmonic priors, which implies there is not much change in *Sruthi* for both baseline and finetuned. Third, when measured on a per-clip win basis the fine-tuned model wins on roughly half of the clips (49–51 %), which underscores the limitation of small fixed-window cosine distances as perceptual proxies and motivates the more discriminating human listening study.

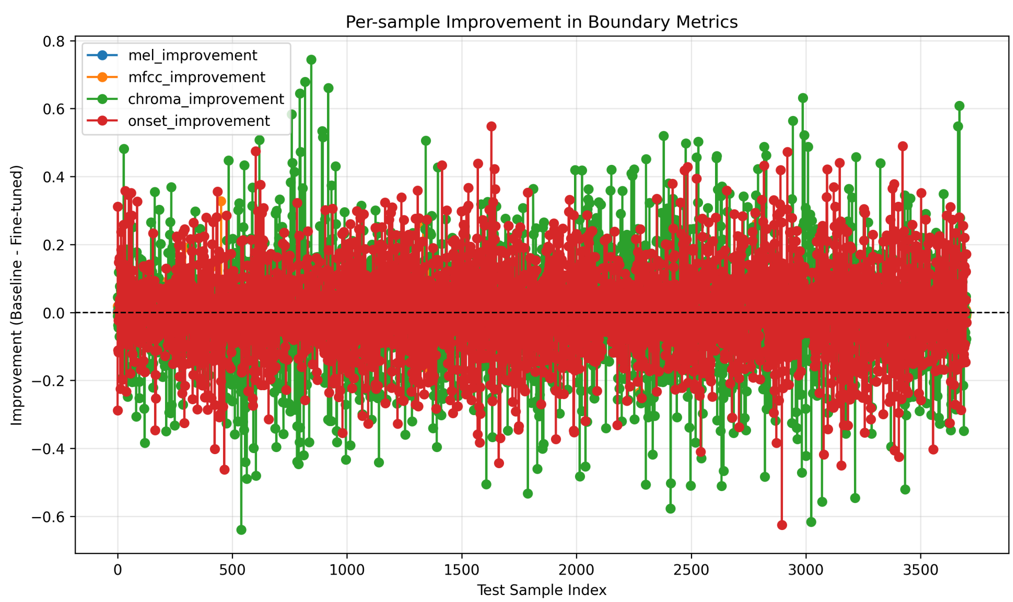

***Figure 5** Per-sample boundary-distance improvement (Baseline – Fine-tuned).*

6.9 Qualitative Evaluation (Human Listening Study)

The qualitative evaluation was deployed as a Streamlit web survey. The survey is structured into a participant-profile stage (Carnatic familiarity Yes/No, formal training level, self-rated knowledge on a 1–5 scale and two consent checkboxes), a 6-trial finetuned model rating and a 6-trial baseline v/s finetuned selection.

In the finetuned model rating section, each trial presents a single 20-second fine-tuned continuation drawn from a balanced mix of performance types (vocal, violin, percussion-mridangam/ tabla) and asks the participant to rate it on three 5-point scales: Musicality, Continuity (smoothness of continuation) and Carnatic style authenticity. In the baseline v/s finetuned selection, each trial presents two unlabeled 20-second clips: one from the baseline model, one from the fine-tuned model, both drawn from the same prompt; and clip order (A vs B) is randomized per participant. The participant is asked three forced-choice questions: which clip has better Musicality, which sounds more authentically Carnatic, and which has smoother continuation. Responses (including a hashed participant ID, response timestamps, trial ordering and free-text comments) are recorded into a Google Spreadsheet via the Google Drive API.

6.9.1 Listening-Study Participant Profile

The Streamlit study collected complete responses from 32 distinct participants. Of these, 15 reported familiarities with Carnatic music and 17 reported no familiarity. Among the familiar group, formal training was distributed across Beginner (6), Intermediate (6), Advanced (7) and Professional (2), with the remainder None.

6.9.2 Finetuned model rating distribution: Musicality, Continuity and Carnatic Authenticity

Across 192 valid trials the fine-tuned continuations clustered around the upper-middle of the 5-point scale, with Carnatic Authenticity attracting the highest mean rating. Figure 7 displays the stacked-bar distribution, and Table 7 reports the means and standard deviations both overall and stratified by listener persona.

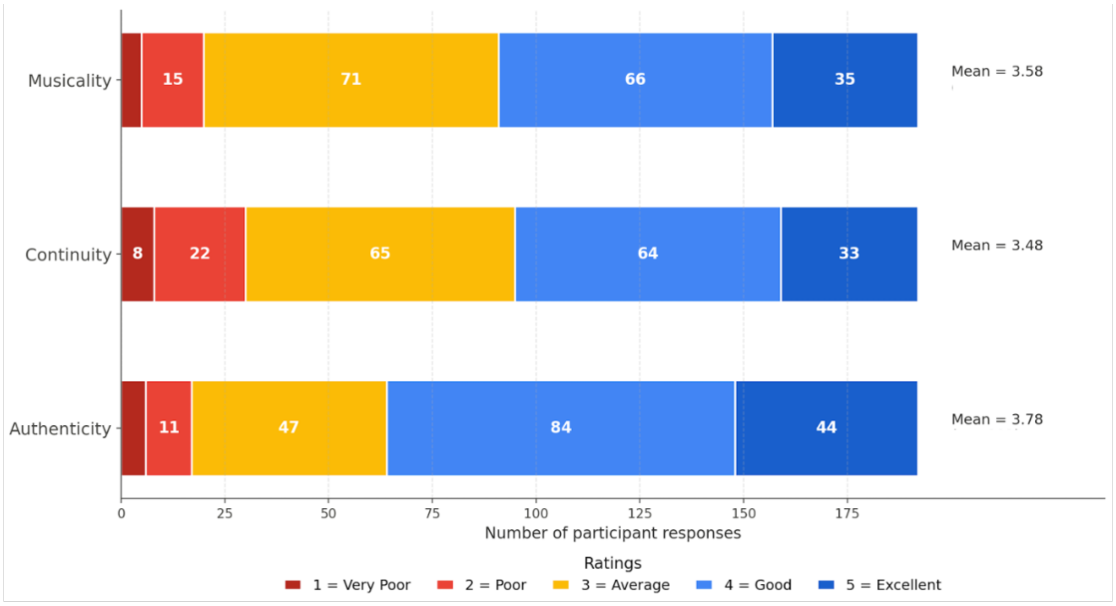

Figure 6 *Distribution of 5-point Likert ratings for the fine-tuned continuations*

A subtle but important pattern emerges from the listener category stratification. Carnatic-familiar listeners gave systematically lower means on Musicality (3.48 vs 3.67) and Continuity (3.41 vs 3.54) than unfamiliar listeners but produced essentially identical means on Carnatic Authenticity (3.79 vs 3.77). This suggests that informed listeners apply a stricter standard to the general musical attributes and can hear shortcomings that unfamiliar listeners do not. But still recognize the Carnatic stylistic markers the fine-tuned model has learned.

Group (n trials)	Musicality (mean ± sd)	Continuity (mean ± sd)	Authenticity (mean ± sd)
Overall (n = 192)	3.58 ± 0.96	3.48 ± 1.04	3.78 ± 0.97
Knows Carnatic (n = 90)	3.48 ± 0.81	3.41 ± 0.94	3.79 ± 0.96
Doesn’t know Carnatic (n = 102)	3.67 ± 1.07	3.54 ± 1.12	3.77 ± 0.98

Table 8 *Mean Opinion Scores (1–5 Likert) for the fine-tuned model, overall and by listener category.*

6.9.3 Finetuned model rating - Means by Performer Type

Stratifying the same 192 trials by performance type (vocal, violin, percussion- mridangam/ tabla) reveals a striking secondary pattern: percussion clips received the highest ratings across all three perceptual axes (Musicality 4.03, Continuity 3.97, Carnatic Authenticity 3.91), violin clips a moderate average and vocal clips the lowest. This is consistent with the difficulty hierarchy of the underlying signal: percussion has a more locally repetitive rhythmic envelope that the model can continue smoothly, whereas vocal and violin alapana involves long-range pitch trajectories and gamaka inflections that are intrinsically harder to maintain. Figure 9 visualizes this performance-type effect.

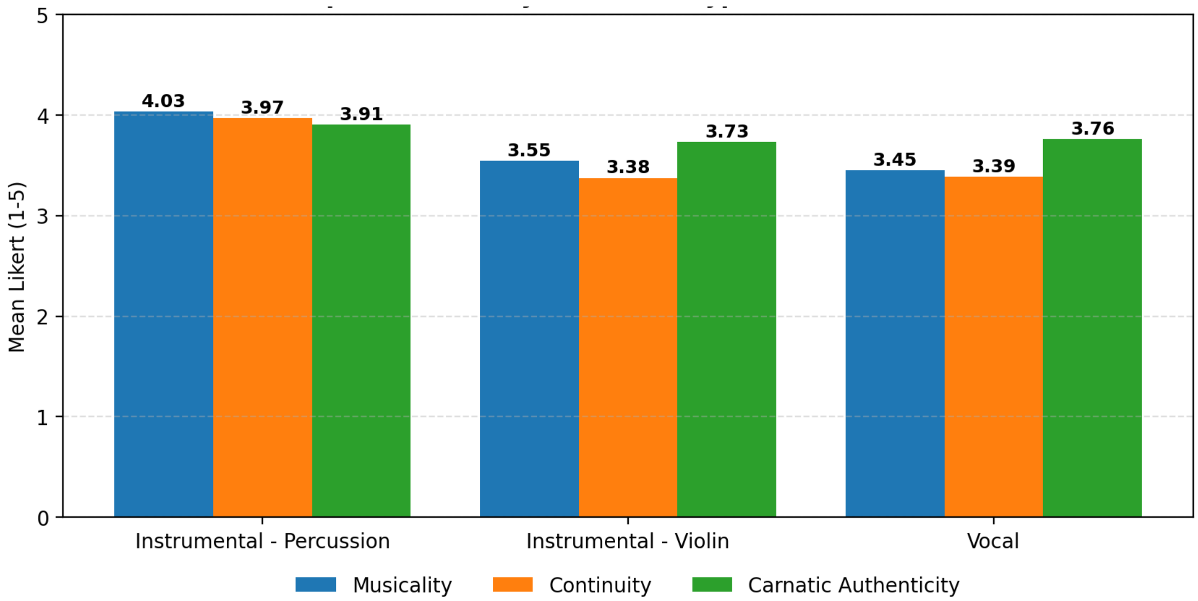

Figure 7 *Mean opinion scores by performance type for the fine-tuned model.*

6.9.4 Preference (Baseline vs Fine-tuned)

Across 192 forced-choice comparisons per question, listeners preferred the fine-tuned continuation in 121 of 192 trials on Musicality (63.0 %), 124 of 192 on Carnatic Authenticity (64.6 %) and 120 of 192 on Continuity (62.5 %). Figure 8 reproduces the headline pairwise chart and Table 9 report the underlying counts.

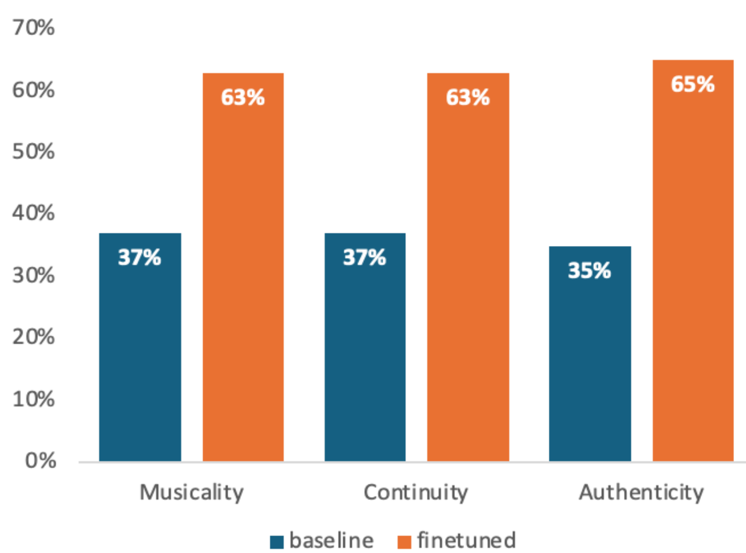

Figure 8 *Headline pairwise preference*

Pairwise question	Baseline chosen	Fine-tuned chosen	Fine-tuned %	n total
Better Musicality (overall)	71	121	63.0 %	192
More Authentically Carnatic	68	124	64.6 %	192
Smoother Continuation	72	120	62.5 %	192

Table 9 *Pairwise A/B preferences*

6.9.5 Preference by Listener Category

When stratified by category, the preference margin is markedly larger among Carnatic-familiar listeners than among unfamiliar listeners. Carnatic-familiar listeners preferred the fine-tuned continuation 68.9 % of the time on Musicality, 72.2 % on Carnatic Authenticity and 68.9 % on Continuity, whereas the corresponding numbers for unfamiliar listeners were 57.8 %, 57.8 % and 56.9 %. This listener category-conditional widening is visualized in Figure 9 and is consistent with the hypothesis that the LoRA fine-tune principally improves Carnatic-specific stylistic markers (raga shape, gamaka envelope, instrument timbre coherence) that informed listeners are best positioned to detect.

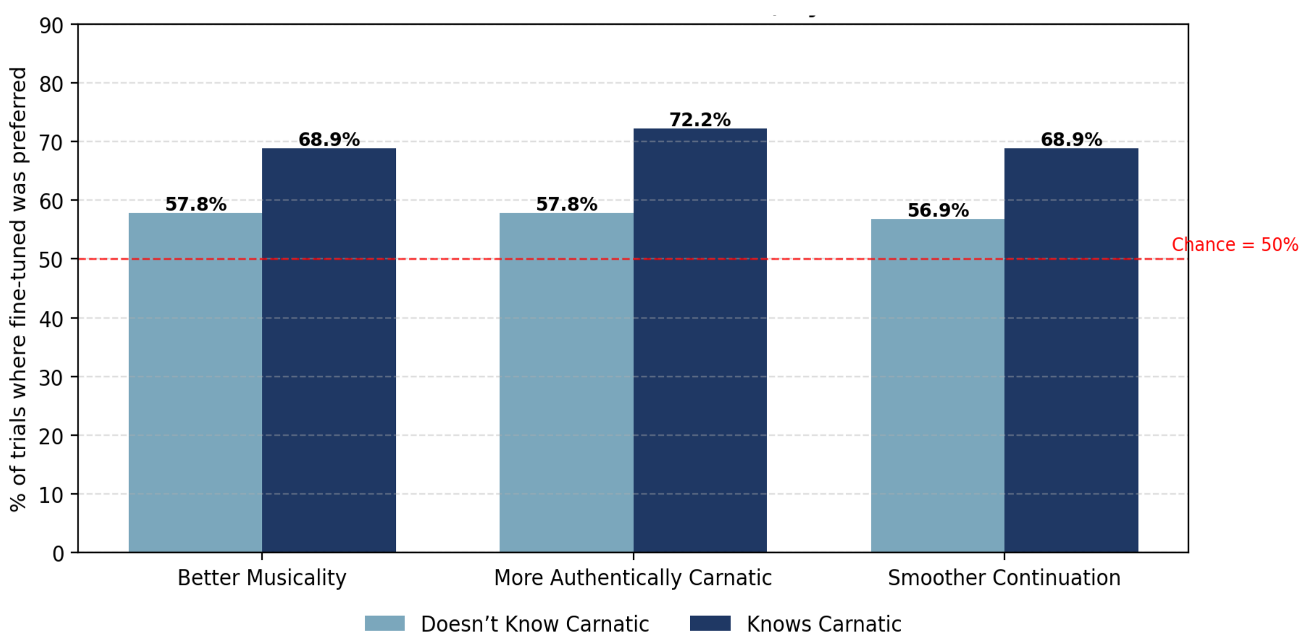

Figure 9 *Finetuned model preference % - stratified by listener category.*

6.9.6 Preference by Performance type

When the same 192 forced-choice trials are stratified by performance type: vocal, violin, and percussion (mridangam), the preference for the fine-tuned model varies sharply across instruments. The LoRA fine-tune improves the model most where the pretrained baseline started furthest from authentic Carnatic style. Performance-conditional fine-tuned-preferred percentages, computed as FT / (Baseline + FT) within each performer type, are reported in Table 10.

Performer type	Better Musicality (FT %)	More Authentically Carnatic (FT %)	Smoother Continuity (FT %)
Violin	**82.40%**	**82.40%**	**78.80%**
Percussion — Mridangam	68.75%	68.75%	70.60%
Vocal	48.00%	52.00%	48.00%

Table 10 *Finetuned model preference % - stratified by performance type*

6.9.7 Statistical Significance Testing

To confirm that the pairwise preferences are not the product of chance, one-sided binomial tests were applied to each of the three pairwise questions under the null H_0: $p \leq 0.5$ versus H_1: $p > 0.5$, where p is the probability that a randomly drawn listener prefers the fine-tuned continuation. Because three tests are performed simultaneously, raw p-values were Holm-corrected to control the family-wise error rate at $\alpha = 0.05$. The results are summarized in Table 11.

Outcome	n	Baseline	Fine-tuned	FT %	Raw p	Holm p	Sig. (α = 0.05)
Better Musicality (overall)	192	71	121	63.02 %	0.000189	0.000378	Yes
More Authentically Carnatic	192	68	124	64.58 %	0.000032	0.000096	Yes
Smoother Continuation	192	72	120	62.50 %	0.000328	0.000378	Yes

Table 11 *One-sided binomial tests with Holm correction.*

All three Holm-corrected p-values are well below the $\alpha = 0.05$ threshold, with the strongest evidence on Carnatic Authenticity (corrected $p \approx 9.6 \times 10^{-5}$). We therefore reject the null hypothesis on every question and conclude that listeners prefer the fine-tuned continuations significantly more often than chance on musicality, Carnatic authenticity and continuity.

The same one-sided binomial test was then repeated within each listener category and within each performer type. In every case, Holm correction was applied across the three pairwise outcomes evaluated for that subgroup. The two subgroup analyses are reported separately in the following sections.

6.9.8 Sub-group analysis - by listener category

For Carnatic-familiar listeners (n = 90), all three Holm-corrected p-values fall well below $\alpha = 0.05$, with Carnatic Authenticity attaining the strongest evidence (corrected $p \approx 4.4 \times 10^{-5}$). The null hypothesis is decisively rejected on every question. For Carnatic-unfamiliar listeners (n = 102), however, none of the three preferences survives correction at $\alpha = 0.05$; the fine-tuned-preferred share of ≈ 57 % is consistent in direction with the familiar group, but the binomial test does not have the power at this sample size to reject H_0 once family-wise error is controlled.

The implication is unambiguous: the pairwise significance reported in Table 11 is essentially driven by Carnatic-familiar listeners. Unfamiliar listeners lean towards the fine-tuned model, but not strongly enough for the test to declare that lean non-random after multiple-comparison correction. This is the strongest single piece of evidence in the study that the LoRA fine-tune is producing culturally specific changes that informed listeners can detect, rather than generic acoustic improvements that anyone would prefer.

Listener category	Outcome	n	Baseline	Fine-tuned	FT %	Raw p	Holm p	Sig. (α = 0.05)
Knows Carnatic Music	Better Musicality	90	28	62	68.89%	0.000219	0.000438	**Yes**
Knows Carnatic Music	Authentically Carnatic	90	25	65	72.22%	**0.000015**	**0.000044**	**Yes**
Knows Carnatic Music	Smoother Continuation	90	28	62	68.89%	0.000219	0.000438	**Yes**
Doesn't Know Carnatic Music	Better Musicality	102	43	59	57.84%	0.068549	0.205648	No
Doesn't Know Carnatic Music	Authentically Carnatic	102	43	59	57.84%	0.068549	0.205648	No
Doesn't Know Carnatic Music	Smoother Continuation	102	44	58	56.86%	0.098895	0.205648	No

Table 12 *One-sided binomial tests with Holm correction - by listener category*

6.9.9 Sub-group analysis - by performance type

Three patterns emerge: (1) Violin (n = 64) is the strongest case statistically. All three Holm-corrected p-values are effectively zero ($\leq 7 \times 10^{-6}$), indicating that on Carnatic violin continuations the LoRA fine-tune produces an effect well outside what could be attributed to chance. (2)

Mridangam (n = 32) sits at the boundary of significance. Raw p-values of 0.020–0.050 reflect a real and consistent preference (fine-tuned was preferred in 69–72 % of trials), but the smaller sample size combined with three-way Holm correction lifts the corrected p-values past the α = 0.05 threshold (0.060–0.100). The effect is directionally consistent and almost certainly real, but the listening study is under-powered on mridangam alone to formally reject H_0 after correction. (3) Vocal (n = 96) shows no detectable preference in either direction. Fine-tuned-preferred shares hover around 47–52 %, raw p-values approach unity, and after correction every test is non-significant. This is consistent with the interpretation in Section 6.9.6 on vocal continuations the pretrained MusicGen-small baseline is already approximately as plausible as the fine-tune, so no significant pairwise preference emerges in either direction.

Performance type	**Outcome**	**n**	**Baseline**	**Fine-tuned**	**FT %**	**Raw p**	**Holm p**	**Sig. (α = 0.05)**
Violin	Better Musicality	64	11	53	82.81%	$< 1 \times 10^{-6}$	$< 1 \times 10^{-6}$	**Yes**
Violin	Authentically Carnatic	64	11	53	82.81%	$< 1 \times 10^{-6}$	$< 1 \times 10^{-6}$	**Yes**
Violin	Smoother Continuation	64	14	50	78.12%	0.000007	0.000007	**Yes**
Percussion – Mridangam	Better Musicality	32	10	22	68.75%	0.050102	0.100204	No
Percussion – Mridangam	Authentically Carnatic	32	10	22	68.75%	0.050102	0.100204	No
Percussion – Mridangam	Smoother Continuation	32	9	23	71.88%	0.020062	0.060186	No
Vocal	Better Musicality	96	50	46	47.92%	0.759649	1	No
Vocal	Authentically Carnatic	96	47	49	51.04%	0.918778	1	No
Vocal	Smoother Continuation	96	49	47	48.96%	0.918778	1	No

Table 13 *One-sided binomial tests with Holm correction - by performance type*

6.9.10 Synthesis across the three significance tests

The three sets of binomial tests: overall (Table 11), by listener category (Table 12) and by performer type (Table 13), together describe a coherent picture. The fine-tuned model produces a statistically significant Carnatic stylistic gain on the whole listener pool view. When the pool is sliced, that gain is concentrated in two specific subpopulations: Carnatic-familiar listeners, who are best equipped to perceive the cultural-style markers the LoRA adapter has learned, and violin (and, directionally, mridangam) continuations, where the pretrained baseline started furthest from authentic Carnatic style and therefore had the most room to improve. On vocal continuations and among Carnatic-unfamiliar listeners the effect attenuates below the significance threshold and pointing to the specific axes (vocal-trajectory modelling, larger adapter capacity, longer continuation horizon) that future work should target.

6.10 Synthesis: Quantitative + Qualitative

The quantitative and qualitative results converge on a single coherent conclusion that a small, parameter-efficient LoRA adapter shifts the MusicGen-small in the direction of Carnatic style, without text supervision. Validation cross-entropy and perplexity decrease monotonically across ten epochs confirming RQ1. On boundary-continuation distance metrics, the fine-tuned model produces small but consistent reductions on three of four features confirming RQ2 partially on Mel, MFCC and Onset that improved, while Chroma remains essentially unchanged. The most discriminating evidence is perceptual: across 192 forced-choice trials per question, listeners prefer the fine-tuned continuation 62.5 %–64.6 % of the time, with all three preferences statistically significant after Holm correction confirming RQ3. The disparity between modest objective gains and larger perceptual gains; and the markedly stronger preferences from Carnatic-familiar listeners; together indicate that the LoRA adapter is shifting culturally salient features that fixed window cosine distances do not directly measure.

7. Challenges & Mitigation

Fine-tuning a Western-trained foundation music model on Carnatic audio surfaced a recurring set of methodological challenges across dataset structure, codec fidelity, compute, evaluation validity and study-design bias. These also appear consistently across recent surveys of foundation music models. This study surfaced its own version of each, anticipated most of them, and applied a corresponding mitigation in the pipeline. Table 14 consolidates the major foreseen challenges; their impact and the mitigations applied in the study. Each row identifies a challenge that was tractable within the study's scope and the corresponding mitigation that was applied.

Where a challenge admitted only a partial mitigation, the residual constraint is carried forward to Section 8 as a documented limitation and to Section 9 as an explicit future-work direction.

Challenge	Impact	Mitigation applied
Long full-length recordings vs. fixed-length training expectation of MusicGen	Poor continuation learning, training instability if raw long audio is fed in	Built a deterministic 10-s segmentation pipeline; recording-level train/val/test split with fixed seed = 42 to avoid leakage.
Recording variability - different microphones, halls, loudness	Reduced model robustness and generalization	32-kHz mono resample for all clips, silence trimming, RMS / zero-ratio / non-silent-ratio quality filters.
Microtonal ornamentation (gamakas) potentially smoothed by EnCodec quantization	Loss of cultural authenticity, melodic fidelity	Used 32-kHz EnCodec (the highest-fidelity variant exposed by MusicGen-small) and validated codec reconstruction informally in stage 2 of the pipeline; future work targets finetuning EnCodec Tokenizer and pitch-contour-aware metrics.
Limited Carnatic dataset size vs. MusicGen pretraining scale	Overfitting, catastrophic forgetting	Adopted LoRA (≈ 0.5 % trainable parameters) instead of full fine-tuning; small LR (1e-4); weight decay (1e-4); early stopping with patience = 3.
Compute / memory limits on Apple Silicon laptop	Training infeasibility, long iteration times	Pre-cached EnCodec tokens once; gradient accumulation 8 to simulate batch 8; CPU continuation generation for determinism.
Objective metrics may not reflect Carnatic perceptual authenticity	Evaluation validity / research credibility	Adopted hybrid evaluation: training CE/perplexity + boundary cosine distances (Mel/MFCC/Chroma/Onset) + listening study (Likert + pairwise) + Holm-corrected binomial tests.
Listening-study design bias (priming, fatigue, inconsistent playback)	Reliability of qualitative evaluation	Streamlit UI with randomized clip ordering per participant, short 6-Likert + 6-pairwise comparison sessions, listener category stratification at analysis time.
Reproducibility risk from undocumented preprocessing decisions	Inability for third parties to replicate results	All thresholds and seeds in lined in helpers.py; per-stage CSVs (segments, splits, tokens, training history, boundary metrics); README and initial_environment_setup.md documenting Apple Silicon environment build.

Table 14 *Foreseen challenges and the mitigations applied.*

8. Unmitigated challenges and limitations

Several constraints could not be eliminated within the scope of this study. They are documented here both for honest disclosure and to seed the future-work agenda.

8.1 FAD-style distributional evaluation could not be performed in its canonical form

The Fréchet Audio Distance (FAD) and other distributional metrics in their canonical formulations require a reference distribution drawn from the same musical context as the generated audio. In a continuation setting, the natural reference for each prompt is the real continuation of that prompt. However, the cleaning and segmentation pipeline aggressively trimmed leading and trailing silence and split each recording into independent 10-second clips before any prompt was selected; the original recordings are not re-loaded at evaluation time.

Consequently, the prompts used for evaluation do not have a corresponding ground-truth continuation tail in the dataset against which generated continuations could be compared distributionally. The same constraint applies to KL-style and embedding-similarity metrics that presume aligned ground-truth pairs. As a result, this study reports cosine boundary-continuation distances (a per-sample local proxy) and Holm-corrected human preferences (a perceptual signal) instead of FAD. Two future-work directions for restoring distributional metrics: (1) Re-segmenting with overlapping prompt + reference-continuation pairs preserved, or (2) Computing an in-domain FAD against the entire test-segment embedding distribution are discussed.

8.2 Per-clip wins on boundary metrics are near 50 %

Although mean Mel/MFCC/Onset distances decrease, the fine-tuned model wins on only ~ 50–51 % of individual clips. This indicates that boundary cosine distances over 1-second windows are a noisy per-sample estimator. Improvements aggregate across the corpus but do not dominate at clip resolution. Stronger objective metrics (multi-scale spectrograms, learned audio embeddings, raga-aware pitch metrics) would be needed to corroborate the perceptual gains at clip resolution.

8.3 Compute and model-size constraints

All training and inference were performed on a single Apple Silicon GPU via MPS, which limited the study to MusicGen-small (the smallest variant) with effective batch size 8 (1 × 8 gradient accumulation). MusicGen-medium and MusicGen-large would likely benefit from greater representational capacity, larger LoRA rank, more number of epochs and more aggressive regularization, but those configurations were not feasible within the available compute budget.

8.4 Microtonal ornamentation through EnCodec

EnCodec, MusicGen's tokeniser, was trained predominantly on Western audio. Although informal codec reconstruction tests on the dataset suggested usable fidelity, fine pitch movements characteristic of gamakas may be partially smoothed during quantization. A higher-bitrate compression model or a domain-aware tokenizer would be a natural improvement direction; pitch-contour-aware evaluation metrics would similarly be a more sensitive evaluation target.

8.5 Listening-study scope and continuation horizon

The study collected 32 participants and analyzed 192 valid Likert and 192 valid pairwise unit responses. While this is sufficient for highly significant binomial tests on the pairwise outcomes, a larger and more demographically balanced cohort with more Advanced and Professional Carnatic musicians would enable subgroup analyses with greater statistical power. Continuations were also limited to a 10-second horizon (20-second total clip with 10-second prompt). Long-form generation, where Carnatic structural concerns such as alapana to kriti to tani transitions become salient, was outside this study's scope and remains an important open evaluation question.

9. Future work

The limitations in section 8 translate into a structured agenda for follow-on work.

- Re-segment the dataset with overlapping prompt to reference-continuation pairs preserved, enabling canonical FAD, KL-divergence and embedding-similarity metrics in addition to boundary cosine distances.
- Scale up to MusicGen-medium and MusicGen-large with larger LoRA rank, longer training and mixed-precision optimization on accelerator hardware (CUDA / TPU) and compare adapter quality across model sizes.
- Fine-tune the EnCodec tokenizer on Carnatic audio so that the input tokens themselves become Carnatic-aware. Either retrain only the residual vector quantizer's codebook entries on the Carnatic corpus (codebook-only fine-tuning) or co-train EnCodec's encoder, quantizer and decoder end-to-end with reconstruction, commitment and adversarial losses, then re-tokenize the corpus and re-run LoRA on the MusicGen LM against the new tokens.

- Adopt Carnatic-aware objective metrics like pitch-class distribution similarity over cents (mod 1200), gamaka extent and modulation index, raga-conditional embedding distances, and multi-scale spectral metrics.
- Investigate longer-horizon continuation (60 s and minute-scale generation), evaluating how alapana shape, kriti development and tala cycles are preserved over time.
- Expand the listening study with a larger and more demographically balanced cohort, enriched by structured musician-only sessions, and use Bayesian hierarchical models to estimate listener category - conditional preference probabilities with proper uncertainty quantification.
- Explore complementary fine-tuning paradigms: (i) Adopt an RLHF-style pipeline analogous to MusicRL[39] to train a Carnatic preference model from listening-study pairwise judgments and use it as a reward signal to further fine-tune the LoRA-adapted MusicGen. (ii) Combine LoRA with prompt-tuning, where a short learned soft-prompt sequence is prepended to the input to provide an additional Carnatic steering signal alongside the LoRA weight updates.

10. Ethical considerations

This study does not collect personally identifying financial or biometric information. The Indian Art Music Raga Recognition Dataset is licensed for research-only use through Zenodo and is neither redistributed nor scraped. The Streamlit listening study presented an explicit informed-consent step before any audio was played and recorded only optional name / phone fields together with non-sensitive musical-background metadata; participation was voluntary and participants could exit at any point. Generated audio is consistently labelled as AI-generated to avoid misrepresentation as authentic performance, and the study does not provide voice or artist cloning capability. The system makes no claim to substitute for human Carnatic artistry or pedagogy, and it is a controlled methodological study of cultural domain adaptation in foundation music models.

11. Reproducibility and code repository

All code, configurations and metadata required to reproduce the pipeline are organized in the GitHub repository. The repository is structured around four orchestration layers namely classes.py (datasets and LoRA module), helpers.py (preprocessing, tokenization, training and evaluation primitives), pipelines.py (high-level stage runners) and fine-tuning.ipynb (the executable notebook chaining all stages). Random seeds are fixed at every non-deterministic boundary (split = 42), preprocessing thresholds and LoRA hyperparameters are recorded inline, and per-stage CSVs (segments_metadata, train/val/test_segments, *_tokens, training_history, boundary_metrics_per_sample) provide a complete audit trail. Initial environment setup, including the AudioCraft fork required for Apple-Silicon builds, is documented in initial_environment_setup.md.

Code repository: github.com/sundarram1608/finetuning-musicgen-small-carnatic-continuation

Listening study (Streamlit): capstone-user-evaluation-survey.streamlit.app

12. Conclusion

This capstone study asked whether a small, parameter-efficient fine-tuning method, LoRA, can adapt an open-source music foundation model like MusicGen-small to meaningfully shift its generative behavior towards Carnatic music using only audio supervision, modest compute and a hybrid evaluation framework. The answer is a qualified yes. LoRA fine-tuning produced monotonic decreases in validation cross-entropy and perplexity, consistent reductions in boundary-continuation distance on three of four acoustic features, and statistically significant pairwise preference for the fine-tuned continuations across all three perceptual dimensions evaluated, with the strongest perceptual margin reported by Carnatic-familiar listeners on the dimension of Carnatic authenticity. The study also documents an honest set of methodological limitations, most notably the unavailability of canonical Fréchet Audio Distance owing to the cleaned segmentation design and translates them into a structured agenda for future work. Taken together, the results contribute a small but concrete piece of evidence that culturally specific domain adaptation of foundation music models is achievable with limited resources, and that audio-to-audio continuation is a viable interface for traditions for which prompt-labelled corpora do not exist.

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